# Keynote: Crypto Agility in the Quantum and AI Era

BGIN Block 15 · Georgetown University Capitol Campus, 111 Massachusetts Avenue NW, Washington, D.C.

| | |
|---|---|
| **Date** | 15 October 2026 |
| **Room** | Room A (2424) |
| **Published time** | 09:20–09:50 |
| **Workstream** | pqc-migration |
| **Status** | Draft for working-group review. Source: closed captions. |
| **Rule** | Chatham House Rule. Information may be used without revealing the identity or affiliation of speakers or other participants, directly or indirectly. |

This is a session record, not a transcript. Substantive discussion is retained with uncertainty and disagreement made explicit. Reported claims are not verified findings. Candidate agreements require working-group confirmation before use as decisions. Actions omit personal owners and identifying roles.

## Agreements

The following sentences stand on their own. These are candidate agreements and reported points of convergence for working-group review. Inclusion does not establish consensus, verify a technical claim, or record formal adoption. The working group must confirm each decision before it is cited as such. Claims that were only discussed, including figures and alternatives that were not adopted, remain in full under Discussion.

Crypto agility — redundancy across assumptions, with a fallback — is the principle that should drive standards, products, and key management.

Standards have to be implementable in real blockchain and payment systems. ECDSA-like "good at everything" performance will not return.

Hybrid constructions (classical plus post-quantum), as in the NIST guidance and in deployments such as X-Wing, are the near-term hedge while post-quantum schemes have short tenure.

Lattice-based and hash-based NIST standards are the two families with the most collective confidence, and neither is assumed permanent.

Signatures are the first primitive to migrate, because signing is control of funds. They are not the whole migration.

The first CRQCs are expected to be slow (days or weeks on ion or neutral-atom machines). Speed comes later and widens the attack from reused keys to in-flight signatures.

Industry will resist agility because it costs more. The cost was treated as unavoidable.

Asset-manager concern is real enough that protocol roadmaps are already being written against it.

## Documents considered

The session referred to the following documents, repositories, and standards. Bibliographic identifications and factual checks added during review are marked as editorial clarifications; they do not imply that the links were spoken or that the meeting adopted the findings. Names of the people who introduced the material are omitted.

NIST post-quantum signature candidates, including HAWK (weakened and withdrawn) and Falcon / FN-DSA (floating-point, not yet standardized in this account).

SLH-DSA as the standardized hash-based signature (SPHINCS).

ML-KEM, named as the scheme people hope will be a sufficient one-time swap.

X-Wing, as a hybrid TLS key-exchange (lattice plus elliptic curve).

RSA, ECDSA, and elliptic-curve pairings / identity-based cryptography, as the classical baseline.

A Google ECDSA resource-estimate paper from the past year. No URL was spoken.

Google and Cloudflare hybrid deployments, as examples, not as speakers.

Clarity Act, as policy context for tokenized U.S. debt and stablecoins.

W3C proposal on crypto agility. No draft name or URL was spoken.

ZK-STARKs and hash-function reductions, as an Ethereum roadmap claim under criticism.

No poster was recorded. The LAMPS drafts on one-signature certificates and on composite FN-DSA with LMS were not discussed in this slice.

## Discussion

The following is the substance of the session. It preserves claims, figures, alternatives, and objections. Discussion does not establish a decision. Any candidate agreement requires explicit working-group confirmation.

The keynote's stated punchline is crypto agility: cryptography going forward has to assume that any one algorithm may fail, and systems have to be built so that failure is survivable. Standards and research were urged to stay inside the practical constraints of the products that will actually ship. A stated failure mode in blockchain work is research that never meets implementation.

Three trends were argued to shape cryptography:

1. **Blockchains as financial rails.** Public-key cryptography is not in the background of the product; keys are part of the user-visible system, so key management is both more important and harder than in traditional finance. The efficiency claim: lower cost and lower friction make these rails likely to carry a large part of future finance. Illustrations used: banks issuing tokenized debt and equities; stablecoins already in the billions to hundreds of billions of dollars of market cap and used across finance. Charts cited for the last couple of years: stablecoin volume from 3.9 trillion to 12.6 trillion, described as evidence of a role in global payment flows, with the growth story outside the United States, where traditional transaction costs have been high because of weak trust. Stablecoin volume was described as growing about 40% year on year, and political fragmentation as likely to enlarge that role. Tokenized

assets were put at about 30 billion dollars, small beside global debt and equity markets, and expected to grow. A macro claim: Western governments are heavily indebted and need new buyers; issuing debt on blockchains, including stablecoins backed by U.S. debt, widens the buyer pool beyond banks and institutional capital. The Clarity Act was cited as policy context for that U.S. debt-distribution motive.

2. **Quantum computing.** Discussed since the 1980s and 1990s. Shor's algorithm, given a large enough quantum computer, breaks the asymmetric algorithms in use today, specifically RSA and ECDSA. A Google paper from the past year, focused on ECDSA rather than on asymmetric cryptography in general, was described as massively reducing the resource estimate for a cryptographically relevant quantum computer (CRQC). That result was said to pull the operational problem forward. The practical claim: the cryptographic systems blockchains use today may be the first systems to break. Academic cryptography rests on math; operational trust also rests on a Lindy effect — hardness assumptions that have survived on the order of 30 years. Post-quantum candidates mostly have much less tenure, so implicit trust is lower, at the same time as the third trend.

3. **AI.** Math breakthroughs are increasing in number and significance, and the math is the foundation of the cryptosystems that would secure blockchain value. A figure given for value secured by blockchains: about 2.7 trillion dollars, varying by the day, with a possible future share of the global economy large enough that weak assumptions are unacceptable.

**Crypto agility, as defined here.** Redundancy inside a critical system. Cryptography becomes a critical system of finance. Agility means not staking that system on one algorithm: if one hardness assumption fails, another assumption is still standing.

**Exposure of current chains, under a slow first CRQC.** Bitcoin, the largest asset by market cap, uses addresses that are hashes of public keys. Value at a Bitcoin address that has been in use since roughly 2014 was described as not vulnerable to a quantum attack until the key is exposed. Reuse — the same address twice, signing twice, or sending twice from the same address — exposes the key. About 32% of all bitcoin was described as exposed to a CRQC. Ethereum addresses function more like identities: interaction with a smart contract registers the address. Address hygiene is therefore much harder. About 65% of Ethereum was estimated to have exposed public keys, on a market cap of roughly 300 billion dollars, varying by the day. Solana and other high-throughput chains were described as having treated quantum vulnerability as unimportant three or four years ago, when the standing assumption was that a CRQC was always twenty years away. These exposure shares assume the first CRQCs are slow. As machines get faster, the claim was that all of this value is at risk: a signature sitting in the mempool, or otherwise attached to a transaction before finality, could be forged inside a short window.

**Why the timeline feels shorter than it did two years ago.** Modalities. Generation-one machines were described as superconducting, pictured as a dilution-refrigerator "chandelier," energy-intensive and fragile. Qubits are fragile and need continuous error correction. Those systems, built since the 2000s, were said not to have scaled. A quantum algorithm still has to beat the classical baseline on

economics, not only on asymptotics. For cryptanalysis the classical baseline is strong: about thirty years of evidence that very large classical computers do not break these schemes, and that a quantum computer would.

Two newer substrates: trapped ions and neutral atoms. They typically run slower, and the qubits are more stable. Stability reduces error-correction work. A slower logical clock gives the error-correction machinery more time. The cost of that stability: recovering a private key from a Bitcoin public key (early coins were the example) can take days or weeks.

Error correction itself has advanced. Resource estimates for a CRQC were said to have collapsed over the past year, landing at the same time as these modality results and as AI-assisted cryptanalysis.

**AI cryptanalysis of post-quantum candidates.** The discussion used recent cryptanalysis to argue for agility. HAWK was cited as a weakened signature candidate. A second reference sounds like McEliece, an established code-based encryption family; Classic McEliece is a key-encapsulation mechanism, not a signature scheme. The claim that this second scheme had been broken is not substantiated by the material identified for this review. Editorial check: the published AI-assisted results describe HAWK and reduced-round AES, and do not establish a practical break of deployed cryptography. They must not be substituted for the unclear second reference. See `https://www.anthropic.com/research/discovering-cryptographic-weaknesses/`

**No Goldilocks replacement.** ECDSA and elliptic-curve cryptography were described as having been a Goldilocks tool: small, fast signatures; enough algebraic structure for zero-knowledge proofs; pairings for identity-based cryptography. No post-quantum scheme is known to carry all of those properties. The sharp tradeoff named: aggressiveness of the hardness assumption versus performance.

SLH-DSA, the standardized hash-based scheme derived from SPHINCS+, was described as conservative. Hash functions are, on present knowledge, secure against quantum computers. The size claim: on the order of one hundred times larger than ECDSA. That was said to be unusable for Solana and for any payment system at high volume, because the network path that carries that throughput would have to be at least 100Œ larger.

Falcon was described as the smallest of the options under discussion, still several times larger than ECDSA, and under consideration at Solana for that reason ("the best bad option"). It has not been standardized, in this account, because it depends on floating-point arithmetic, and there is no canonical hardware handling of floating-point numbers. The same key can produce different signatures on different platforms, which opens attack surface that is not fully understood.

Bitcoin's design center was described as "digital gold": low transaction rate, longevity of the cryptography as the product. Solana's design center was described as Visa-scale speed, with an advertised figure of about 50,000 transactions per second. Those two design centers do not pick the same post-quantum signature.

The same split was extended to national standards. Internet cryptographic standards were described as having been set in a post-Cold War window. The present claim: China intends to compete with the United States; Russia and the United States are antagonistic; many other states do not want their

cryptography tied to any one of those powers. Payment systems will have to carry national standards and decentralized-chain standards at the same time. That Balkanization is the operational case for agility.

**What "we don't know" was said to require of an architecture built in 2026 and expected to last.** Modular by default, because the algorithm set is not known per use case. Able to switch. Able to sign a blockchain transaction under one or more algorithms. Confidence in deployed post-quantum schemes is low because they have secured value for a short time. Hybrid construction was described as the NIST direction and as what Google and Cloudflare are doing. X-Wing was used as an example of hybrid key establishment. Editorial clarification: X-Wing combines ML-KEM-768 and X25519; it is a key-encapsulation mechanism, not a signature scheme. The discussion's redundancy argument concerns combining assumptions; its cost comparison was not a measured benchmark. Specification: `https://www.ietf.org/archive/id/draft-connolly-cfrg-xwing-kem-08.html`

Policy was placed under the cryptography. Elliptic curves were cheap enough to deploy everywhere and let applications sit on top. The prompt offered to the standard-setting sessions: decide what the system is for, derive the policy, and only then derive the cryptographic requirements.

The same pattern was claimed for any high-assurance signing and key-management flow, not only digital assets. Agentic finance needs runtime controls: cryptographic policy that limits agents acting for people or institutions, and that policy has to be agile. Physical AI (robotics and autonomous systems) was illustrated with military drones: the software-update path from the manufacturer is a critical authenticity pipeline, and a compromise of an armed autonomous platform is severe. Insurance and underwriting were cited as cases where a failure of authenticity can cost billions. High-assurance authenticity is where agility matters most. High-volume, low-value flows can accept weaker tradeoffs. Goals should drive both products and standards.

A closing technical caveat: a useful quantum computer would also be a scientific instrument (a check on quantum physics; possible uses in medicine and astronomy) and a possible cryptographic tool. Quantum random-number generation and quantum key distribution were cited as schemes whose assumption is a physical property rather than an unsolved math problem. They were described as impractical today and as worth watching. The phrase used was that they could be an "ultimate" cryptography because the assumption is physical.

The closing handoff introduced the BGIN PQC migration competition and referenced the Google Quantum AI resource-estimate paper and associated proof material. ECDSA.fail, an Eigen Labs quantum-circuit optimization project, was a separate benchmark reference. It is also distinct from sig.golf, the signature-optimization challenge discussed later. Public project attribution does not identify any meeting contributor. Competition details are developed in the first PQC migration report.

**Questions, in substance.** How many signature-algorithm turnovers should be expected? A number was refused. The reframed question was the lifespan of any one algorithm. Quantum arrival was described as a step change: once Shor's algorithm can be run, today's elliptic-curve targets break. Other quantum algorithms might later break other assumptions; that development would be long-running.

AI is a separate unknown for the underlying math. The working assumption: future cryptosystems have much shorter lives than RSA and ECDSA have had. Searching for one Goldilocks replacement was called a dangerous frame. The hope offered was short-term stability across a few assumptions and tradeoffs, with agility as the reason that shorter lives are tolerable.

Agility implies continuous migration. A project was said to have been proposed at W3C about two months before the meeting, to standardize movement from one cryptographic paradigm to another. It was described as not yet having taken hold. The work in the room was not described as part of that W3C project. The W3C effort was still credited with helping define an open parameter: how fast an algorithm may be deployed (a day, an hour) against the security purpose. The architectural claim: stop making the cryptographic assumption the immovable foundation of a platform. Treat policy and authorization as the stable layer, and cryptography as a lifecycle component.

Why systems were not built agile from the start. No objection was expected at the level of the idea. The objection is engineering cost: complexity raises the cost to implement and to maintain. That cost was offered as the reason industry has been slow, despite years of proposals from W3C working groups and researchers. The cost was also described as unavoidable if value and information are to keep moving. TLS version migration was used as lived evidence that a new standard does not propagate by itself. Treating a swap to ML-KEM as enough to "go back to normal" was called a dangerous assumption that people will nonetheless make. The Y2K comparison was rejected: Y2K was a finish line; this is a standing regime.

Whether concrete policy can be written in advance. A concrete foundation was called the wrong metaphor, because ironclad faith in one foundation will not be available. The metaphor offered was an immune system: layered defenses, a fallback when one layer fails, and adaptation as the threat changes. Algorithm choice was described as heuristic: what has survived longest. Lattices were called comparatively strong. A custody design under discussion uses both lattice-based and hash-based NIST standards, the two families with the most collective confidence, and does not assume either family is permanent.

What "agile" means once cost, speed, and accuracy are in tension. The answer was tied to the use case and to the assurance level required. The extreme agile design — every client signs with every cryptosystem ever proposed, classical or post-quantum — was called the highest bar and impractical, with nuclear launch codes as the only example where it might be justified. For payments, one hybrid combination may be enough.

What "PQC migration is finished" means if it depends on the use case. The talk's center was signatures for authentication and authorization: the ability to sign for a wallet is ownership of the funds. The Ethereum roadmap was described as also requiring privacy in a post-quantum setting, which adds systems beyond signatures. A claim associated with that roadmap — that security can be reduced to a hash function, and that hash functions are known to be secure, so the reduction is the strongest available cryptography — was called optimistic. ZK-STARKs were the concrete counterexample: promoters describe them as "just hashes," but a large amount of machinery sits on the hash and has to be evaluated on its own. Open conjectures on proximity, in code-based systems that ZK-STARKs also

use, were said to leave the implication unclosed. A signature assumption and a privacy assumption were not treated as interchangeable. One primitive does not close every dimension of the migration.

How much quantum fear is already changing institutional Bitcoin exposure. It was called a significant barrier to gaining exposure that would otherwise be taken. Evidence that asset managers are actively short the asset because of quantum risk was called limited. Ethereum, Solana, and Zcash were described as positioning a post-quantum roadmap, and therefore longevity, against ecosystems whose plan is less structured; Bitcoin was the comparison, sometimes unnamed. That positioning was described as a response to asset-manager hesitation. The sequence offered: asset managers recognized the theft risk first, about two years ago, because they would bear the loss; protocol foundations followed; custodians and other infrastructure that serve those asset managers have, with limited exceptions, treated the question at the surface.

A custody architecture was offered as a case study of the modular, switchable, multi-algorithm design above, aimed at high-assurance signing for digital-asset custody. A post-quantum testnet for Ethereum was described as under construction, with the aim of positioning that network. Design partners mentioned as categories: a privacy-preserving layer 1 (proper name garbled), a stablecoin payment system, and several foundations as early partners. Incumbent custody was described as built on elliptic curves, including hardware roots of trust and multi-party computation that both depend on today's cryptography, so the migration is of the surrounding systems as well as of the algorithm.

### Matters not resolved

One replacement algorithm versus a standing agile regime with shorter algorithm lifetimes. The single-replacement view, including "ML-KEM and we are done," was rejected in the room's framing.

Whether Y2K is a usable analogy. It was rejected.

How many algorithm generations will be needed. Left open; lifespan was the preferred question, and no number was given.

How concrete a policy can be before the threat is known. Layered, adaptive policy was preferred to a fixed foundation. The assurance level still depends on the use case (payments versus extreme high assurance).

Whether "everything reduces to a hash, so recursive ZK-STARKs are secure" follows. It was disputed. Proximity conjectures on the code-based components were left open. Privacy and signature security were not accepted as one assumption.

How exposed Ethereum is, chain by chain, once CRQCs are fast rather than slow. The 32% / ~65% figures are for a slow first machine and for exposed keys, not for a fast mempool attack on all value.

Which post-quantum schemes survive AI-assisted cryptanalysis. HAWK was discussed as weakened; the apparent McEliece reference and claimed attack remain unverified. These are not two confirmed breaks of signature schemes.

Falcon versus SLH-DSA is a live design split (Solana-scale throughput versus Bitcoin-scale conservatism), not a settled choice. Floating-point Falcon was treated as incompletely understood.

Whether quantum risk is already priced as selling pressure. Concern and delayed exposure: yes. Active shorting: thin evidence.

A W3C crypto-agility standardization proposal exists as an idea and was said not to have taken hold. Speed-of-change targets (hour, day) are undefined.

National-standard Balkanization was asserted as a design constraint, not as a negotiated outcome.

### Disposition of the work

A modular custody design was presented as the pattern: many algorithms, switching, single- or multi-algorithm signatures on blockchain transactions, lattice plus hash inside the system, policy underneath the cryptography.

An Ethereum post-quantum testnet was described as in progress.

Design-partner categories in hand: a privacy-preserving layer 1, a stablecoin payments system, and foundation clients. Incumbent custody stacks were described as still on elliptic curves.

W3C: a crypto-agility / paradigm-migration proposal about two months old, not yet established. Not part of the work presented.

The slice ends as the PQC-migration competition is announced. Status detail is in the next slice.

### Actions

Each action is written so that it can be tracked without the rest of the record. Where the room named only a person, the action is kept and the name is omitted.

Use the following sessions to define what systems are for, then derive policy, then cryptographic requirements.

Treat algorithm lifespan and continuous migration as the design problem, rather than a count of one-time swaps.

Keep quantum key distribution and quantum random-number generation on a watch list; they were not proposed as 2026 deployment tools.

The competition announcement (resource estimates, testbeds, a standard) is the immediate next block of work.

### Coverage

This record covers the published keynote in Room A. The keynote ran into the first PQC migration session. The published title is used. The speaker is not named. Figures are reported session claims, not independently validated measurements. Technical names are corrected where supported by the recording, chat, or identified primary documentation. Where a name could not be checked against another part of the same recording, it is left unverified rather than guessed. Logistics, introductions, and names of speakers are omitted.

Zoom join, screen-share permission, a microphone check, thanks, and a commercial fundraising question. Hosting pleasantries at the handoff into the PQC session are noise. The competition announcement

itself is substance and is specified in the next slice.

—.

**Source clarifications and open checks**

Apparent McEliece reference: the code-based family is identifiable, but the claimed attack, affected variant, parameters, and security impact require a paper citation. A break of Classic McEliece is not established here.

Privacy-preserving layer-1 partner, rendered "cache foundation" / "Soc.".